

Forecast Figures: What They Show

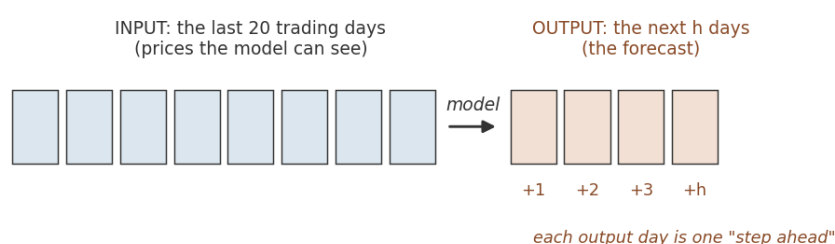
KAN and LSTM one-day to two-hundred-day forecasts on CBA.AX, under the corrected evaluation protocol.

1. How a forecast is produced

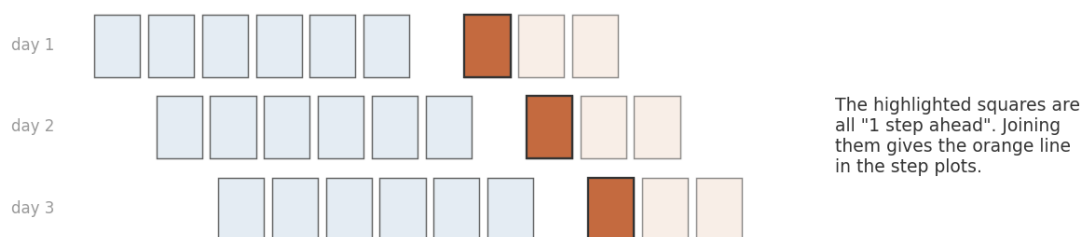
Each model works on a sliding window. It is shown the closing prices of the last 20 trading days, and from those it predicts the next h days in one go, where h is the forecast horizon. So a 100-day model outputs one hundred numbers at once, not a single number.

The window then slides forward one day and the model does it again. Repeating this across the test period produces a great many overlapping forecasts, which is why there is more than one sensible way to draw the result.

A. How a single forecast is made



B. Step plots: fix one step, follow it through time



C. Trajectory plots: fix one date, show its whole path

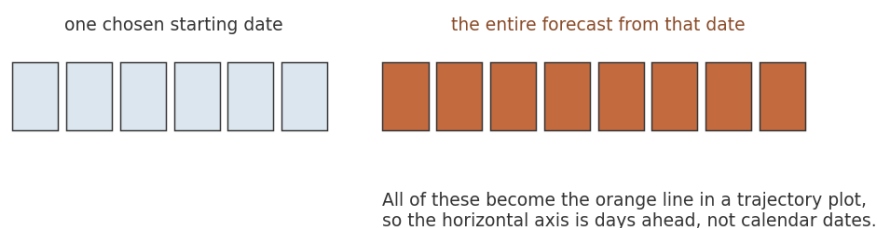


Figure 0. How a forecast is made, and what each style of plot takes from it.

2. The two kinds of plot

Step plots (Figures 1 to 4). Pick one distance into the future, for example one day ahead, and follow only that prediction as the window slides forward. Panel B of the diagram shows this: the highlighted square in each row is the same distance ahead each time, and joining them produces one line. The question answered is: *if the model predicts N days ahead, how well does it do, day after day?*

In a step plot the horizontal axis is **calendar time**, labelled with dates. Each point on the line is a separate forecast, made on a different day.

Trajectory plots (Figures 5 and 6). Pick one date and show the entire forecast made on that date, all h days of it, against what actually happened. Panel C of the diagram shows this. The question answered is: *standing on this particular day, what did the model think the next hundred days would look like?*

In a trajectory plot the horizontal axis is **days ahead**, labelled +1, +2 and so on, not calendar dates. The whole line comes from a single forecast rather than many.

Step plots are the ones to use when comparing models, because they average over many forecasts. Trajectory plots are better for showing how a model behaves when asked to see a long way ahead.

3. What the axes mean in every figure

The vertical axis is always the closing price of the stock, scaled to a range of 0 to 1. The lowest price in the sample period becomes 0, the highest becomes 1, and everything else sits in between. So 0.85 means the price is at 85 percent of the way between the cheapest and dearest days on record for this stock. These are not dollars. The scaling was applied before the models were trained and all published error figures use the same scale, so the numbers are comparable with the paper's tables.

The horizontal axis depends on the plot type, as described above: calendar dates for step plots, days ahead for trajectory plots.

The lines. Black is what actually happened. Blue is the LSTM forecast. Orange is the KAN forecast. Where a coloured line sits on top of the black one, the model was right; where it departs, that gap is the error.

The error figure quoted under each plot is the root mean square error of that particular line: roughly, the typical size of the vertical gap between the coloured line and the black one. Lower is better, and because the prices are scaled, an error of 0.02 means a typical miss of about 2 percent of the stock's full price range.

Every figure uses the final walk-forward test fold, which is the most recent stretch of data, and one the models never saw during training.

1. One day ahead

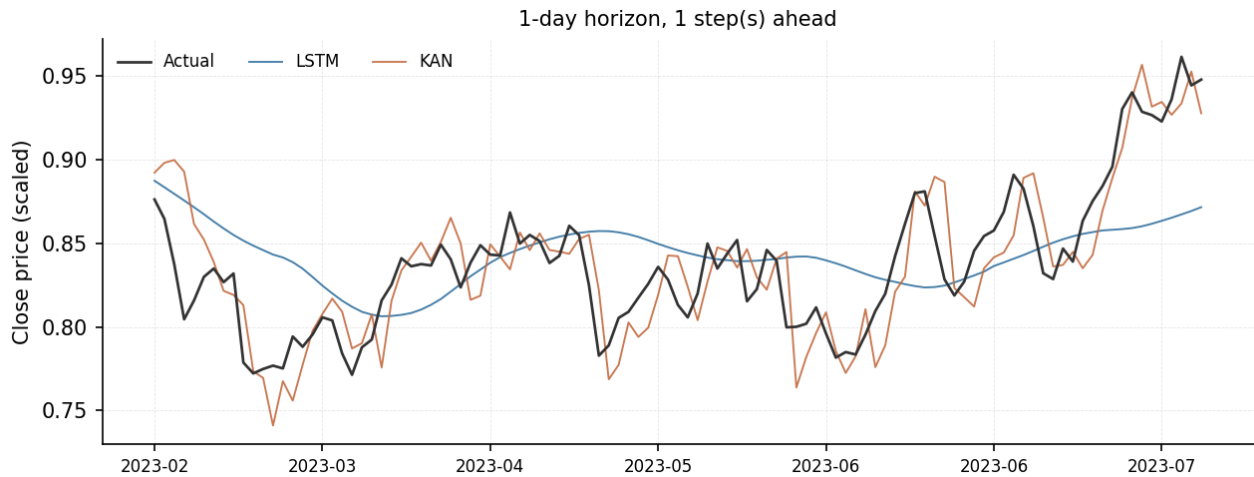


Figure 1. One-day-ahead forecasts against the actual price, February to July 2023.

Horizontal axis: calendar date, from February to July 2023. Each point is a fresh forecast made on that day.

Vertical axis: closing price, scaled from 0 to 1.

Error on this line: KAN 0.024, LSTM 0.035 (lower is better).

Both models stay close to the actual price, which is not surprising, since tomorrow's price is usually near today's. The KAN follows the daily movement more closely. The LSTM is smoother and tends to arrive at each turn slightly late.

2. Two days ahead

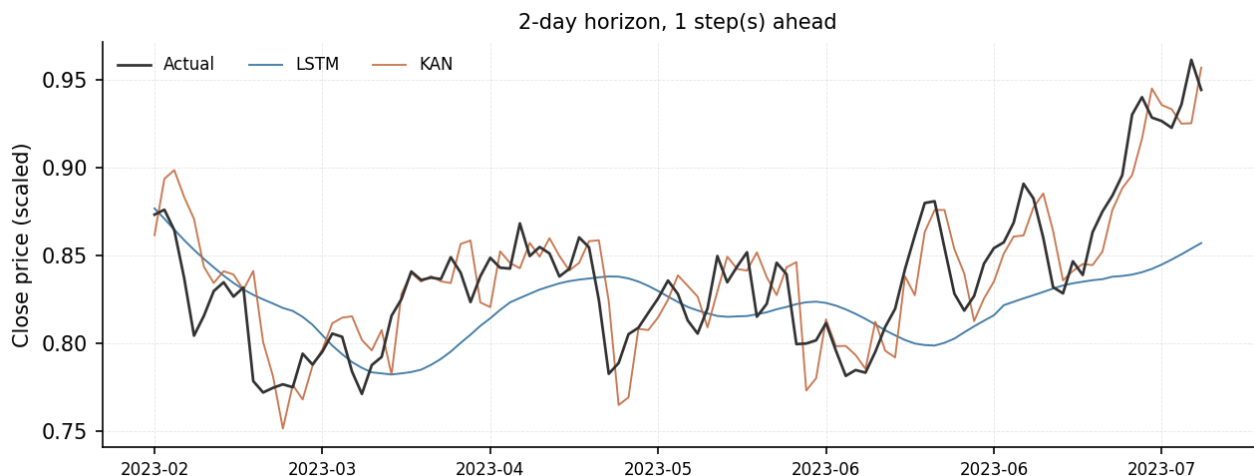


Figure 2. Two-day-ahead forecasts against the actual price.

Horizontal axis: calendar date. Each point is a forecast made two days before the day being predicted.

Vertical axis: closing price, scaled from 0 to 1.

Error on this line: KAN 0.021, LSTM 0.040 (lower is better).

This is the clearest of the six. The KAN tracks the real movement, including the sharp falls in mid February and early June. The LSTM has effectively drawn a smooth average through the data: it captures the general shape but misses almost every turning point, and from late June it drifts well

below the actual price as the market climbs.

If one figure had to show that the two models do not perform as previously reported, this is the one.

3. One hundred days ahead

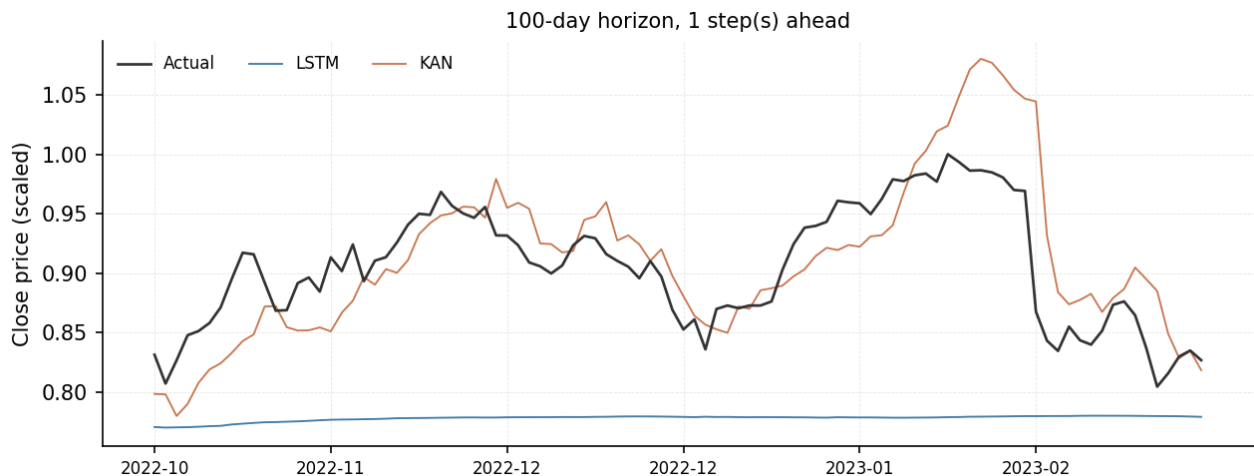


Figure 3. Forecasts made one hundred days in advance.

Horizontal axis: calendar date. Each point is a forecast made one hundred days in advance.

Vertical axis: closing price, scaled from 0 to 1.

Error on this line: KAN 0.043, LSTM 0.137 (lower is better).

The same comparison, but each forecast is now made a hundred days before the event. Both models are visibly further from the truth, and the LSTM sits noticeably above the actual price for much of the period.

4. Two hundred days ahead

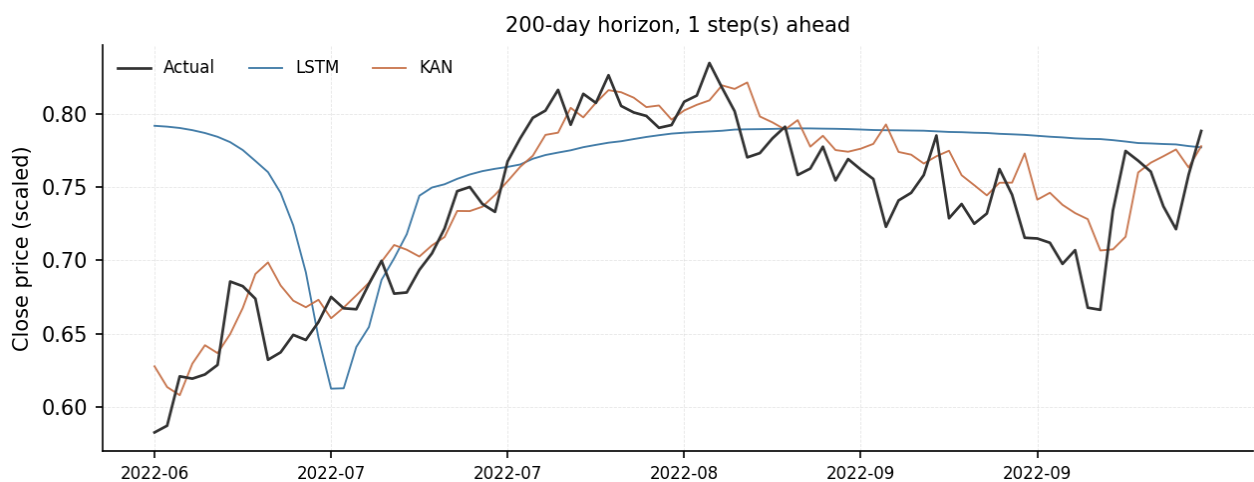


Figure 4. Forecasts made two hundred days in advance.

Horizontal axis: calendar date. Each point is a forecast made two hundred days in advance.

Vertical axis: closing price, scaled from 0 to 1.

Error on this line: KAN 0.026, LSTM 0.068 (lower is better).

At this distance neither model has much real information to work with. Both lines stay within a narrow band while the actual price moves around them.

5. A single hundred-day forecast path

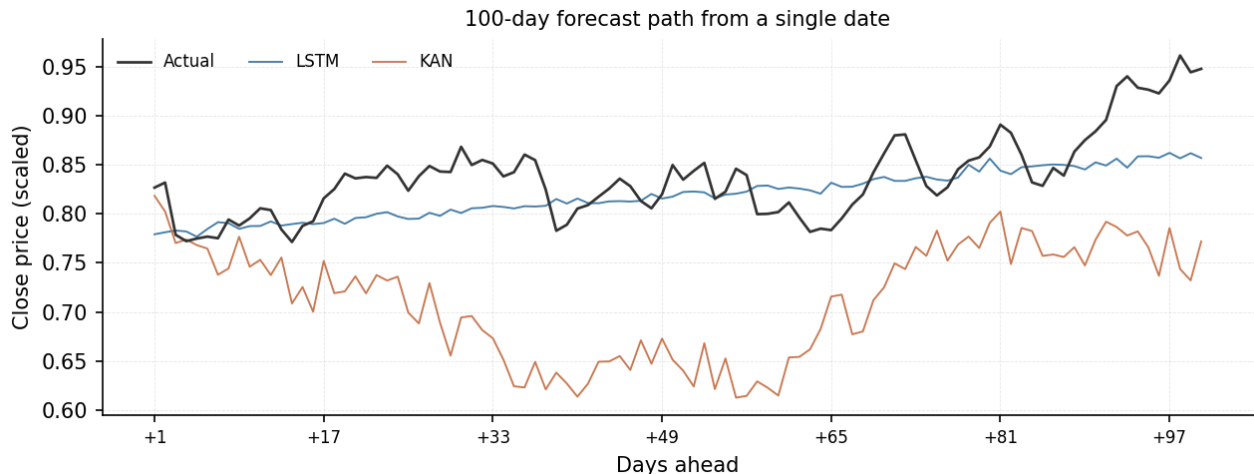


Figure 5. One starting date, and what each model expected over the following hundred days.

Horizontal axis: days ahead of the chosen starting date, from +1 to +100. This is not calendar time.

Vertical axis: closing price, scaled from 0 to 1.

Error on this line: KAN 0.139, LSTM 0.037 (lower is better).

The LSTM produces an almost flat line with a slight upward drift. It has essentially learned the average price level and predicts that, rather than any movement. The KAN swings well below the real price through the middle stretch before recovering.

Two quite different ways of being wrong, and neither is visible in a table of error figures.

6. A single two-hundred-day forecast path

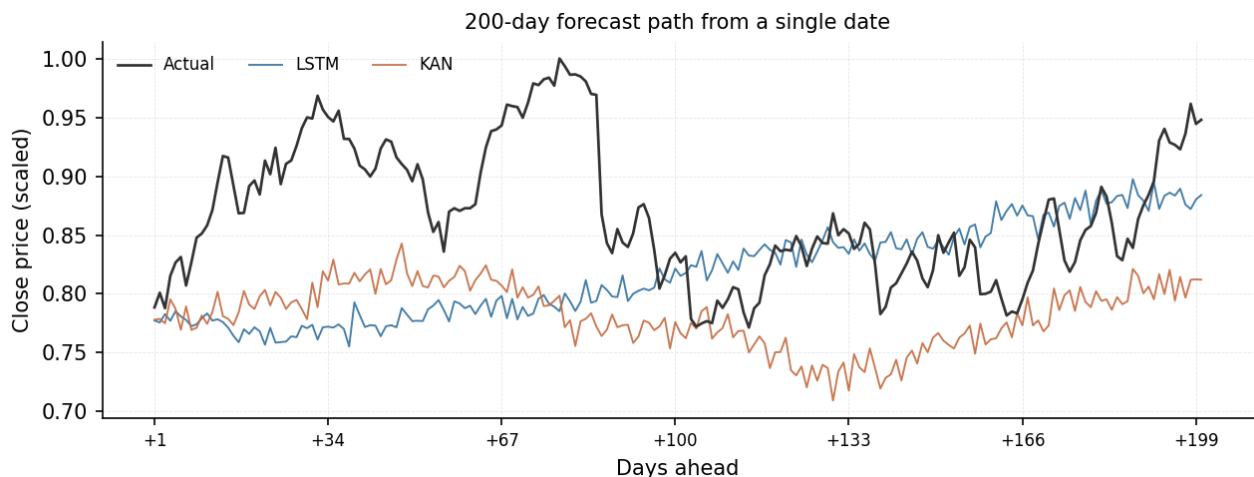


Figure 6. One starting date, and the following two hundred days.

Horizontal axis: days ahead of the chosen starting date, from +1 to +200. This is not calendar time.

Vertical axis: closing price, scaled from 0 to 1.

Error on this line: KAN 0.101, LSTM 0.097 (lower is better).

The most honest picture in the set. The actual price rises to the top of its range around day 75 and swings considerably. Both forecasts stay inside a narrow band for the entire two hundred days, drifting gently.

Neither model is predicting the future here. Both have learned roughly where the price tends to sit, and little else.

The short version

At short horizons the KAN follows real movement more closely, while the LSTM smooths it away. Errors grow with distance for both, as expected. At long horizons neither model forecasts in any meaningful sense: they settle near an average and stay there.

One caution

Figures 5 and 6 each show a single starting date, so they illustrate behaviour rather than prove it. The errors quoted for them apply to that one path and will not match the pooled figures in the paper's tables. Use them to show *how* the models behave, not *how well*.

Regenerating these

All six figures come from `make_figure.py` in the folder above. Change the settings at the top of that file and run it; each run writes a PNG for viewing and a `.tex` version for the paper. Re-run `build_report.py` afterwards to rebuild this document, which recomputes every error figure from the logged runs so the text cannot drift from the images.